

Public Chat Portal for AlKhidmat

Kaavish Report
presented to the academic faculty
by

Syeda Samah Daniyal	sd07838
Fatima Dossa	fd08024
Rubab Shah	rs08104
Syeda Yashfeen Zehra Zaidi	sz08469



In partial fulfillment of the requirements for
Bachelor of Science
Computer Science

Dhanani School of Science and Engineering

Habib University
Spring 2026

Contents

1	Introduction	4
1.1	Problem Statement	4
1.2	Proposed Solution	4
1.3	Intended User	6
1.4	Project gantt chart and deliverables	7
1.5	Key Challenges	9
2	Software Requirement Specification (SRS)	12
2.1	Functional Requirements	12
2.2	Use Cases	19
2.3	Non-functional Requirements	19
2.4	System Diagram	22
2.4.1	Presentation Layer (Client-Side)	23
2.4.2	Application Layer (Backend)	23
2.4.3	AI/ML Layer	24
2.4.4	Data Layer (Storage and Retrieval)	24
2.4.5	Infrastructure Layer (Deployment)	24
2.4.6	Relationships Between Layers	25
2.4.7	Flow of Interaction	25
2.5	External Interfaces	26
2.5.1	User Interfaces - Wireframes	26

List of Figures

1.1	Month-Wise Deliverables	7
2.1	Use Case Diagram of the Public Chat Portal for Alkhidmat	19
2.2	High-level System Architecture	22
2.3	User Interface using Whatsapp	26
2.4	Staff Dashboard	27
2.5	Further Contents in Ticket Content Panel	28
2.6	Further Contents in Ticket Content Panel	28
2.7	Admin's Dashboard - Analytics Monitor	29

List of Tables

2.1	Non-Functional Requirements	20
-----	---------------------------------------	----

1. Introduction

1.1 Problem Statement

Alkhidmat Foundation Pakistan, one of the largest non-profit organizations in the country, handles thousands of daily inquiries from donors, volunteers, and beneficiaries regarding donations, healthcare, and community welfare programs. Currently, these queries are managed manually through phone calls, emails, or in-person visits. This system leads to inefficiencies, delayed responses, redundant questions, and overburdened staff while also alienating overseas or low-literacy users as they struggle to navigate static website content or locate authentic project data.

The lack of an automated communication solution restricts inclusivity and accessibility, especially for marginalized communities or overseas donors seeking quick, verified information. Moreover, Alkhidmat’s operations span multiple domains; healthcare (hospitals, pharmacies, diagnostics, blood banks), education, disaster management, orphan care, women empowerment, and more, creating a vast and complex information network that manual systems cannot efficiently support.

Hence, the problem is twofold:

1. Operational burden due to repetitive, manually handled queries.
2. Limited accessibility for donors and beneficiaries due to language, literacy, and technology barriers.

The motivation behind this project is to bridge this gap by creating an AI-driven, multilingual, multimodal public chat portal that ensures inclusivity, transparency, and efficient information flow which empowers both users and staff.

1.2 Proposed Solution

The proposed solution is an AI-powered Public Chat Portal for Alkhidmat Foundation, a unified platform integrating a Retrieval-Augmented Generation (RAG)-based

LLM with multimodal capabilities (text and image input). The portal will be accessible through Alkhidmat’s official website and WhatsApp Business API to allow seamless communication across all major user groups; donors, patients, blood donors, and beneficiaries.

System Overview:

1. **Frontend Chat Interface:** Accessible through two primary channels:
 - **WhatsApp Interface:** Integrated via the WhatsApp Business API, allowing users to chat directly through WhatsApp using text, voice notes, and image uploads. The interface employs structured message templates and quick-reply buttons for guided workflows, ensuring inclusivity for overseas donors and non-technical users.
 - **Web Interface:** Built in React for responsive and multilingual interaction (English, Urdu, Roman Urdu), providing a modern and intuitive experience for both desktop and mobile users.
2. **Backend Services:** Powered by FastAPI and LangChain, the backend manages query classification, retrieval, and response generation. It includes:
 - **Intent Classification Module:** Determines the user’s domain of inquiry (e.g., Donation, Healthcare, Blood Services, General Support).
 - **Retrieval-Augmented Generation (RAG) Engine:** Combines LLM reasoning with a verified, domain-specific knowledge base to produce contextually accurate and concise responses.
 - **Multimodal Processing Layer:** Handles image understanding and context management for smooth multimodal input handling.
 - **Security and Compliance Layer:** Enforces end-to-end encryption, anonymization of user data, and access control for backend APIs.
3. **Knowledge Base (RAG Datasets):** Curated and periodically updated datasets covering:
 - Donation and payment workflows
 - Healthcare and welfare program details
 - Donor engagement and transparency data
 - CSR and corporate partnership information
 - Accessibility and support resources

4. **Human Escalation Dashboard:** A staff-facing interface where unresolved or sensitive queries are routed to human agents. Staff can respond directly, monitor conversation histories, and update relevant dataset entries.
5. **Analytics and Monitoring Module:** Tracks user engagement, response accuracy, and escalation frequency. Insights from these analytics help refine the system’s knowledge base and improve LLM performance over time.

1.3 Intended User

The chat portal caters to three primary user groups, identified through persona analysis from workshops with Alkhidmat:

1. **Donors (Local, Overseas, Corporate):**

- Seek transparency, convenience, and verified information about donation channels, project impact, and tax compliance.
- Subtypes include recurring donors, new donors, first-time donors, and CSR/corporate donors.

2. **Healthcare Beneficiaries (Patients, Blood Donors, Recipients):**

- Require guidance on healthcare services, test prices, facility locations, doctor availability, ambulance booking, and Zakat-based assistance.
- Includes rural and urban patients with varying digital literacy levels.

3. **General Users (Public Information Seekers):** Represent individuals who approach Alkhidmat for general inquiries such as volunteer opportunities, branch contact details, upcoming drives, disaster relief information, and awareness campaigns. The chatbot provides them with general updates, project overviews, and links to participation or registration forms.

Through intent classification, each user’s domain (donation, healthcare, general inquiry) is automatically detected, ensuring relevant responses and routing. Administrative users (Alkhidmat staff) access a dashboard to manage escalations, update datasets, and broadcast campaigns via WhatsApp APIs.

1.4 Project gantt chart and deliverables

		YEAR 1				YEAR 2				End of project
	Oct 2025	Nov 2025	Dec 2025	Jan 2026	Feb 2026	March 2026	April 2026			
	Requirements approval	Design completion	Prototype ready	Human routing operational	Full integration	Deployed beta	Final defence			
Phase / Task	Phase 1– Project Initiation & Requirements Gathering	System Design & Data Preparation	Prototype Development – Intent Classifier & RAG Setup	Phase 2 – Human Dashboard & Escalation Routing	Integration & Testing	Analytics Module & Cloud Deployment	Final Testing, Documentation & Presentation			
	- Requirements document - Domain map (12 domains) - Alkhidmat User flow & chatbot conversation map	- UX wireframes for chat interface - Dataset collection & cleaning from Alkhidmat resources - Initial ER diagram and API routes	- Trained ML classifier (intent detection) - Integrated RAG pipeline using LangChain + Llama 2 - Basic chatbot interface connected to backend	- React dashboard for staff queries - Escalation logic & workflow integration - Internal test report (AI ↔ Human routing)	- Full system integration (chat portal + dashboard + RAG) - Unit & integration tests - Initial usability feedback	- Analytics dashboard (query trends, escalation rates) - Dockerized deployment (Heroku) - Beta version live for demo	- Final project report & documentation - User guide & demo video - Final FYP presentation and viva			
Responsible Members	All members (collaboration) Lead: Samah (documentation structure), Fatima (domain mapping)	1.Fatima: UX wireframes 2. Samah: Database schema & API design 3. Yasmeen: Data preprocessing 4. Rubab: Data logging plan	1. Yasheen: Classifier model 2.Fatima: Frontend integration 3.Samah: Backend FastAPI endpoints 4.Rubab: Data logs for test runs	1.Samah: Backend logic & Dashboard APIs 2.Rubab: Dashboard UI/UX 3.Fatima: Chatbot connection 4. Yasheen: Confidence-based routing model	1.Samah: Backend & deployment tests 2. Yasheen: Model evaluation 3.Rubab: Usability test reports 4.Fatima: Frontend bug fixes	1.Rubab: Analytics module 2.Samah: Docker + Heroku deployment 3. Fatima: UI enhancements 4. Yasheen: Data validation metrics	All members (joint effort) 1.Samah: Technical report 2.Rubab: Visuals & analytics 3.Fatima: Presentation slides 4. Yasheen: Model performance summary			
Deliverable Output	Project charter, domain map, workflow	UX wireframes, DB schema	Basic AI chatbot working	Dashboard connected	Working AI + Human chat flow	Live Dockerized version	Documentation & presentation			

Figure 1.1: Month-Wise Deliverables

Deliverables Overview

The Gantt chart above outlines the project objectives, key milestones, and timelines for each phase across Kaavish I and Kaavish II. Major deliverables include:

- **Kaavish I:** System requirements analysis, dataset design, RAG knowledge base creation, chatbot workflow design, and prototype implementation of the frontend and backend communication pipeline.
- **Kaavish II:** Integration of AI/ML modules (intent classification, multimodal processing), deployment on cloud infrastructure (Heroku/Docker), human escalation dashboard, and final evaluation with Alkhidmat's stakeholders.

Resources Needed

While the Gantt chart defines tasks, timelines, and team responsibilities, successful execution also depends on the following essential resources:

1. Hardware and Computing Resources

- Developer laptops with minimum 16GB RAM and GPU support for model inference.
- Cloud hosting credits or Heroku dynos for deployment and load testing.
- External SSDs for backup of datasets and logs.

2. Software and Development Tools

- **Backend:** FastAPI, Python, LangChain, and Llama 2 integration.
- **Frontend:** React.js, Tailwind CSS, and WhatsApp Business API integration.
- **Database and Storage:** PostgreSQL, Elasticsearch, and Firebase (for chat logs and persona data).
- **Version Control and CI/CD:** GitHub, Docker, and Heroku pipelines.

3. AI/ML and Data Resources

- Access to pretrained LLMs (e.g., Llama 2 or OpenAI API for evaluation).
- Dataset curation support from Alkhidmat domain experts for healthcare, donation, and welfare programs.

- Preprocessing tools for Roman Urdu and Urdu text normalization.

4. Human Resources

- **Development Team:** 4 members (Frontend, Backend, NLP, and Integration roles).
- **Supervisors:** Internal advisor for technical oversight; external industry mentor from Alkhidmat.
- **Domain Experts:** Staff providing verified data and validating chatbot responses.
- **Test Users:** Donors, patients, and volunteers for pilot testing and feedback collection.

5. Other Resources

- Secure API keys for WhatsApp Business and cloud hosting.
- Access to Alkhidmat’s internal documentation, SOPs, and donation workflow data.
- Dataset annotation tools for custom tagging of healthcare and donation intents.

These resources collectively ensure that each phase of development outlined in the Gantt chart is feasible and aligned with the technical and operational goals of the project.

1.5 Key Challenges

This section mentions the key challenges that we foresee in this project and possible ways to address them.

1. Data Acquisition and Knowledge Base Consolidation (Pre-deployment)

- **Challenge:** Acquiring data and building a reliable RAG knowledge base is an immediate challenge since Al-Khidmat does not currently have a single, consolidated, structured database. This requires extensive web scraping and manual formatting of disparate documents across multiple domains, which is complicated by time constraints and the need for domain expertise.

- **Mitigation:** This challenge has been explicitly communicated to the external supervisor. The team has prioritized the most impactful datasets through user personas (e.g., Blood Donation SOPs). The mitigation strategy involves assigning a dedicated industry partner resource to gather and provide this essential, formatted data ASAP, aiming for the core datasets to be compiled within the next 1–2 weeks.

2. Multilingual and Roman Urdu Accuracy (Natural Language Processing)

- **Challenge:** Achieving high accuracy in Urdu and handling the inconsistency of Roman Urdu (Urdu written in English script) is difficult, as LLMs often lack high fidelity for low-resource languages and phonetic variations. This directly impacts the $\geq 90\%$ accuracy goal.
- **Mitigation:** Dedicate significant effort to curating the Urdu component of the RAG knowledge base. Employ custom preprocessing steps to normalize common Roman Urdu spellings before feeding them to the LLM/RAG engine.

3. RAG Data Volatility and Latency (Data Management)

- **Challenge:** Al-Khidmat’s program details (e.g., active disaster relief efforts, current blood stock) change frequently. Since the system relies on a **periodically updated** knowledge base, there is a risk of providing outdated information, which compromises user trust and data integrity.
- **Mitigation:** Establish clear and enforced SLA for Admin updates to the RAG knowledge base (e.g., daily update requirement for time-sensitive domains like Blood Services). Use the Human Escalation Dashboard to flag and correct critical data mismatches immediately.

4. Balancing Transparency and Privacy (Security and Ethics)

- **Challenge:** Donors seek transparency about fund utilization, but the system must rigorously protect beneficiary and personal data (e.g., Zakat recipient details, patient health information) while maintaining local compliance standards.
- **Mitigation:** Implement stringent anonymization/pseudonymization for all data logs. Use guardrails to strictly prevent the LLM from generating or retrieving specific identifiable information in public responses.

5. Human Handoff and Agent Training (Operational/UX)

- **Challenge:** Ensuring a seamless handover from AI to a human agent without frustrating the user, particularly on WhatsApp where the 24-hour window constraint applies. Agents also need specialized training to use the Human Escalation Dashboard efficiently (diagnosing the bot's failure, not just the user's query).
- **Mitigation:** Focus on quick context loading and pre-filled response templates in the dashboard. Develop mandatory training modules for agents specifically on "AI Failure Diagnosis" and persona-aware communication.

2. Software Requirement Specification (SRS)

This chapter provides detailed specifications of the system under development.

2.1 Functional Requirements

Functional requirements are grouped into modules corresponding to major system components.

- Core Chatbot Module (Public Facing)
 - Multimodal Input Handling
 - * Text Input: Accept text queries in English, Urdu, and Roman Urdu (basic level).
 - * Image Input: Analyze uploaded images for contextual information (e.g., payment receipts, prescriptions, donation screenshots). Handwritten text recognition is out of scope for MVP.
 - * Voice Input (Phase 2 - Conditional): Convert voice messages to text using Speech-to-Text API. Basic Urdu support only; dialect variations are out of scope. Voice integration depends on successful testing of Phase 1 text-based system.
 - *MVP Scope: Phase 1 focuses on duo-lingual text (English + Urdu) and image input only.*
 - Intent Classification & Domain Identification
 - * Domain Classification: Automatically identify the query domain from four categories:
 - Donation: Payment methods, project details, Zakat/Sadaqah eligibility

- Healthcare: Hospital locations, services, appointments, ambulances
- Blood Services: Donor eligibility, blood bank locations, blood group availability
- General Inquiry: Organization information, volunteering, events, queries about other domains
- *Note: this does not cover the exhaustive list of data for the domains mentioned*
- * Intent Clarification Loop: When queries are ambiguous or confidence score is low, prompt user with clarifying questions before proceeding (e.g., "Are you asking about blood donation or healthcare services?").
- * Confidence Threshold: Queries with confidence score below 0.7 are flagged for human escalation.
- Contextual Response Generation
 - * RAG-Based Retrieval: Retrieve verified information from the domain-structured knowledge base using Retrieval-Augmented Generation.
 - * Persona-Aware Responses: Adapt response length, tone, and detail based on identified user persona (e.g., overseas donor, non-tech-savvy patient, recurring donor).
 - * Follow-Up Context: Maintain session context to handle follow-up questions that depend on prior conversation history.
 - * Session Management: Preserve conversation context throughout the user session to enable multi-turn dialogues.
- Multilingual Response & Accessibility
 - * Language Selection: Allow users to select their preferred language (English or Urdu) at the start of interaction.
 - * Quick Reply Options: Provide button-based shortcuts and pre-defined options for low-literacy users to minimize typing.
 - * Consistent Quality: Ensure response quality is consistent across languages with no discrepancy between English and Urdu outputs.
- Trust & Verification
 - * Branding Display: Show verified Alkhidmat branding, official logos, and chatbot verification messages to assure users of authenticity.
 - * Official Channel Verification: Display verified WhatsApp Business number and official website domain to prevent scam concerns.

- Guardrails & Safety
 - * Content Moderation: Implement guardrails to prevent inappropriate responses or off-topic conversations.
 - * Privacy Protection: Ensure no personal data is retained beyond the session lifetime.
 - * Fallback Responses: Provide predefined safe responses when the system cannot confidently answer a query.
- Information Retrieval & Data Management Module

The knowledge base contains periodically updated, domain-structured information covering (note: this is not an exhaustive list of the data for each domain):

 - RAG Knowledge Base (General Information)

Note: Knowledge base is NOT real-time; it is periodically updated by administrators.

 - * Donation Support:
 - Active and verified campaign details (healthcare, education, orphan care, women empowerment, disaster relief)
 - Payment methods and step-by-step guidance (Easypaisa, JazzCash, bank transfer, international cards, PayPal)
 - Zakat and Sadaqah eligibility criteria for different programs
 - Donation workflow and confirmation processes
 - Transparency reports and impact summaries
 - * Healthcare Information:
 - List of Alkhidmat hospitals, clinics, labs, and pharmacies with addresses and coordinates
 - Service offerings per facility (OPD, diagnostics, pharmacy, ambulance)
 - Operating hours and appointment procedures
 - Consultation fees, test charges, and medicine pricing
 - Doctor availability, specializations, and gender-specific staff information
 - Emergency helpline numbers and ambulance booking procedures
 - * Program Information:
 - Summaries of Alkhidmat’s 12+ domains (education, water, orphan care, community services, disaster management)

- Women-centric initiatives (Bano Qabil, maternity services, women empowerment)
- Youth and student programs
- Corporate Social Responsibility (CSR) partnership opportunities
- * General Information:
 - Organization history, mission, and credibility verification
 - Volunteering opportunities and community event schedules
 - FAQs on trust, transparency, and fund utilization
- Blood Donation Microservice (Operational Data)

A separate microservice containing transactional and inventory information integrates with the main system to handle blood donation queries:

 - * Donor Eligibility Validation: Check eligibility based on age, weight, hemoglobin levels, blood pressure, and donation intervals (WHO and Alkhidmat internal SOPs).
 - * Blood Bank Locations: Retrieve nearest Alkhidmat-licensed blood bank addresses and contact information based on user location.
 - * Blood Group Availability: Query current blood stock levels and urgent blood group requirements.
 - * Donation Process Information: Provide step-by-step donation process details (registration, screening, donation, post-care instructions).
 - * Donor Network Registration: Allow users to register for SMS/WhatsApp alerts when their blood group is urgently needed.
 - * Post-Donation Guidelines: Provide medical instructions (hydration, rest, when to resume activities).
 - * Recipient Support: Guide blood recipients on required documents (CNIC, prescription), cross-match duration, reservation policies, and service charges.
- User Profiling & Persona Management (Contextual Database)

Maintain a separate database to store user information and enable personalized interactions:

 - * Persona Classification: Classify users into personas based on interaction patterns using a donor-beneficiary matrix:
 - Donor personas: Overseas, Local, Recurring, First-Time, Corporate, Non-tech Savvy

- Healthcare personas: Rural/Urban Patient, New Patient, Non-tech Savvy Patient
 - Blood service personas: New/Recurring Blood Donor, Blood Recipient
- * Interaction History Tracking: Record user query patterns, preferred language, interaction frequency, and topics of interest.
- * Dynamic Persona Updates: Update user persona classification after each interaction based on new behavioural data.
- * Context-Aware Response Adaptation: Tailor response length, explanation depth, and recommendations based on identified persona (e.g., provide detailed explanations for first-time donors, concise responses for recurring donors).
- * Personalized Recommendations: Suggest relevant campaigns, services, or programs based on user history and persona.
- Human Agent Dashboard (Staff Facing)
 - Escalation & Routing
 - * Automatic Escalation: Forward queries with confidence score below 0.7 to relevant department-based chat agents.
 - * Domain-Based Routing: Route escalated queries to specialized agents (Donation team, Healthcare team, Blood Services team, General Support).
 - * Sensitive Query Handling: Automatically escalate queries involving complaints, refunds, or personal grievances to human agents.
 - Conversation Management
 - * Real-Time Takeover: Allow human agents to join ongoing conversations within the same chat thread without disrupting user experience.
 - * 24-Hour WhatsApp Window: Enable agents to respond within the 24-hour WhatsApp Business API interaction window.
 - * Pre-Approved Templates: Provide agents with Meta-approved message templates for initiating conversations outside the 24-hour window.
 - * Conversation History Access: Enable agents to view full conversation context before responding.
 - Knowledge Base & Dialogue Management

- * Content Updates: Allow administrators to add, modify, or archive FAQs and program information in the RAG knowledge base.
 - * Dialogue Flow Definition: Enable admins to define and update pre-approved greeting messages, fallback responses, and guided conversation flows.
 - * Guardrail Configuration: Provide admin interface to set and adjust content moderation rules and response boundaries.
- Role-Based Access Control
 - * Chat Agent Access: Grant domain-specific agents access only to queries within their department.
 - * Admin Privileges: Provide full system access to administrators for knowledge base management, guardrail settings, and user management.
 - * Audit Trail: Log all agent actions (view, respond, resolve) with timestamps for accountability.
- Analytics & Logging Module
 - Usage Tracking
 - * Query Metrics: Record total queries, query volume by domain, peak usage times, and average session duration.
 - * Response Time Monitoring: Track response time per query (target: 3-15 seconds based on complexity).
 - * Escalation Rate Tracking: Monitor percentage of queries escalated to human agents by domain.
 - User Feedback Collection
 - * Post-Interaction Rating: Prompt users for thumbs-up/thumbs-down rating 15-20 minutes after their last query.
 - * Follow-Up Query Check: Send a message asking if the user has any remaining questions before requesting feedback.
 - * Satisfaction Score Calculation: Aggregate user satisfaction metrics over time.
 - Performance Analysis
 - * Answer Accuracy Monitoring: Track percentage of queries resolved by AI without escalation (target: $\geq 90\%$).

- * Domain Performance: Analyze which domains have highest accuracy, longest response times, or most escalations.
 - * Trend Identification: Highlight common query topics and identify emerging information needs that require knowledge base updates.
- Reporting & Insights
 - * Dashboard Visualization: Provide real-time analytics dashboard for administrators showing KPIs and trends.
 - * Export Functionality: Generate CSV/PDF reports of usage metrics for management review.
 - * Anomaly Detection: Flag unusual query patterns (e.g., spam, repeated failures, sudden traffic spikes).
- Out of Scope
 - Handling of regional Urdu dialects is beyond the current scope. The system is designed to support only standard Urdu syntax and semantics.
 - The chatbot does not process handwritten text to extract text from images.
 - Support for additional regional languages such as Punjabi, Sindhi, Pashto, or Balochi is out of scope. The MVP exclusively supports Urdu and English text input.
 - The chatbot does not produce voice-based or audio-generated replies. All communication remains text-based for now.
 - The LLM does not store personal data beyond basic identifiers (name, contact number, email). If such data already exists in the Alkhidmat database, it is retrieved for continuity; otherwise, the LLM treats the user as new. No long-term memory or profiling is maintained.
 - Real-time data access is out of scope. The LLM retrieves information from the existing database, which requires manual updates. If the database is not updated, the system will rely on previously stored information and cannot refresh data automatically.
 - Following features are out of scope as for phase 1 of development; In case of a successful phase 1 completion, these will be added to the scope.
 - * Bilingual speech input support for chatbot interaction, enabling automatic transcription and intent detection from spoken English or Urdu.

- * Enhanced understanding and generation of Roman Urdu text for more natural and localized user communication.
- * Advanced WhatsApp broadcasting and personalized campaign messaging to improve outreach effectiveness.

2.2 Use Cases

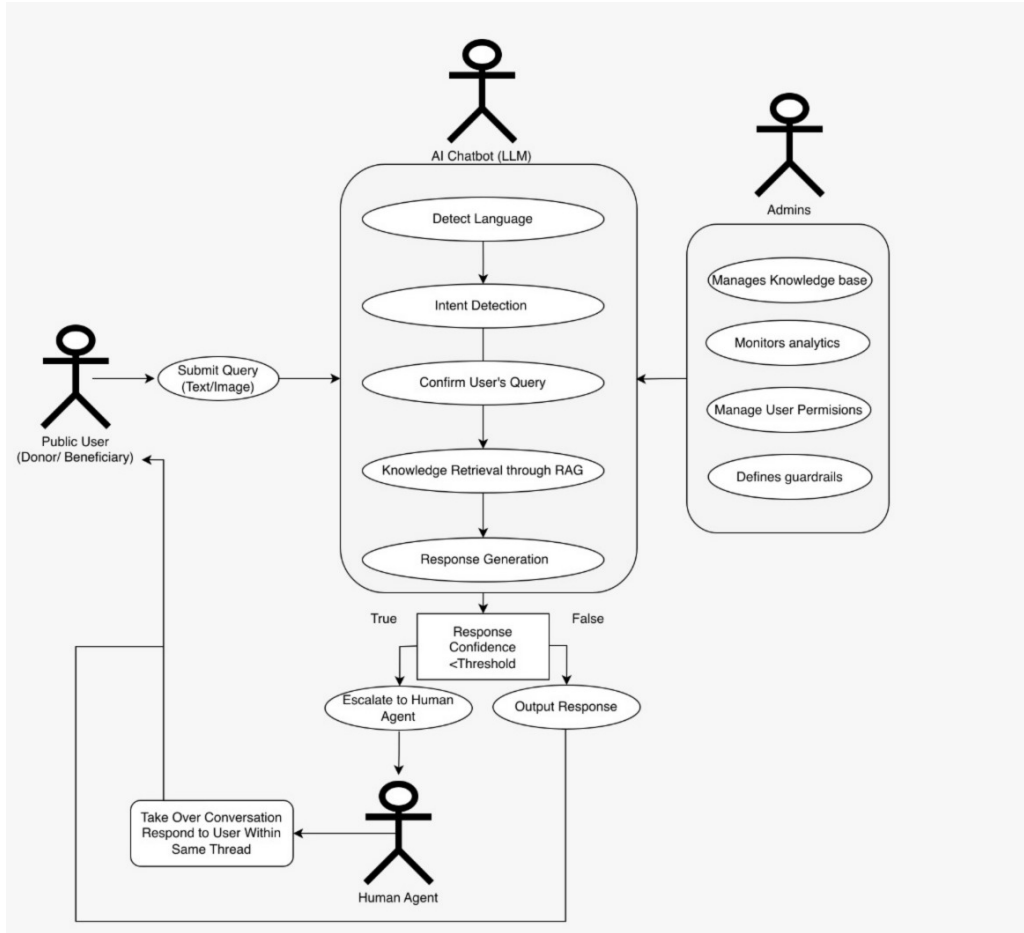


Figure 2.1: Use Case Diagram of the Public Chat Portal for Alkhidmat

2.3 Non-functional Requirements

Table 2.1: Non-Functional Requirements

Category	Description	Example
Performance	The system should respond within 3 to 15 seconds per query, depending on the complexity.	Simple queries (e.g., blood bank locations) should return results in 3 seconds. Complex queries (e.g., eligibility criteria) may take up to 15 seconds.
Accuracy and Relevance	Ensure the system provides $\geq 90\%$ accuracy and relevance in its responses to ensure reliability.	If a user asks for blood donation instructions, the response should be factually correct and relevant in 90% of cases.
Usability / Accessibility	Support multilingual responses with consistent quality in both languages (e.g., English and Urdu). Include clear button-based flows for non-technical users.	The system should provide responses in Urdu with the same clarity and accuracy as in English, without discrepancies.
Reliability	Ensure the system maintains $> 90\%$ answer accuracy and minimal human routing. If accuracy falls below a threshold, route to a human agent.	If a user asks for donation details and the system’s confidence is below 70%, it should route to a human agent.
User Satisfaction (KPI)	Measure user satisfaction by asking for feedback 15–20 minutes after the last query with thumbs up/down ratings.	After a session, if the user does not respond or replies with “no,” send a rating question (thumbs up/down) to evaluate the system’s effectiveness.
Data Integrity & Backup	Ensure data accuracy and periodic backups of important data (e.g., RAG knowledge base). Implement automated weekly backups and data validation.	Perform weekly automated backups of key data and ensure that data validation occurs during data entry to prevent corruption.

Continued on next page

Table 2.1 (continued): Non-Functional Requirements

Category	Description	Example
Security / Compliance	Implement end-to-end encryption for all communications. Ensure no personal data retention beyond session lifetime.	Personal data should not be retained after the session ends. The system should use encrypted communication protocols to secure sensitive information.
Scalability	Support ≥ 500 simultaneous users without performance degradation using FastAPI + Elasticsearch on Docker/Heroku.	The system should scale to support 500 simultaneous users accessing the platform without any degradation in response time.
Maintainability	Use modular code architecture with clear documentation and API version control to ensure the system is easy to maintain and update.	The codebase should be modular with proper documentation to allow for easy updates and troubleshooting.
Compliance and Ethics	Adhere to Pakistan’s data privacy guidelines and AI ethics principles for NGO use.	Ensure that the system follows local data privacy regulations and complies with ethical AI practices for non-profit organizations.

2.4 System Diagram

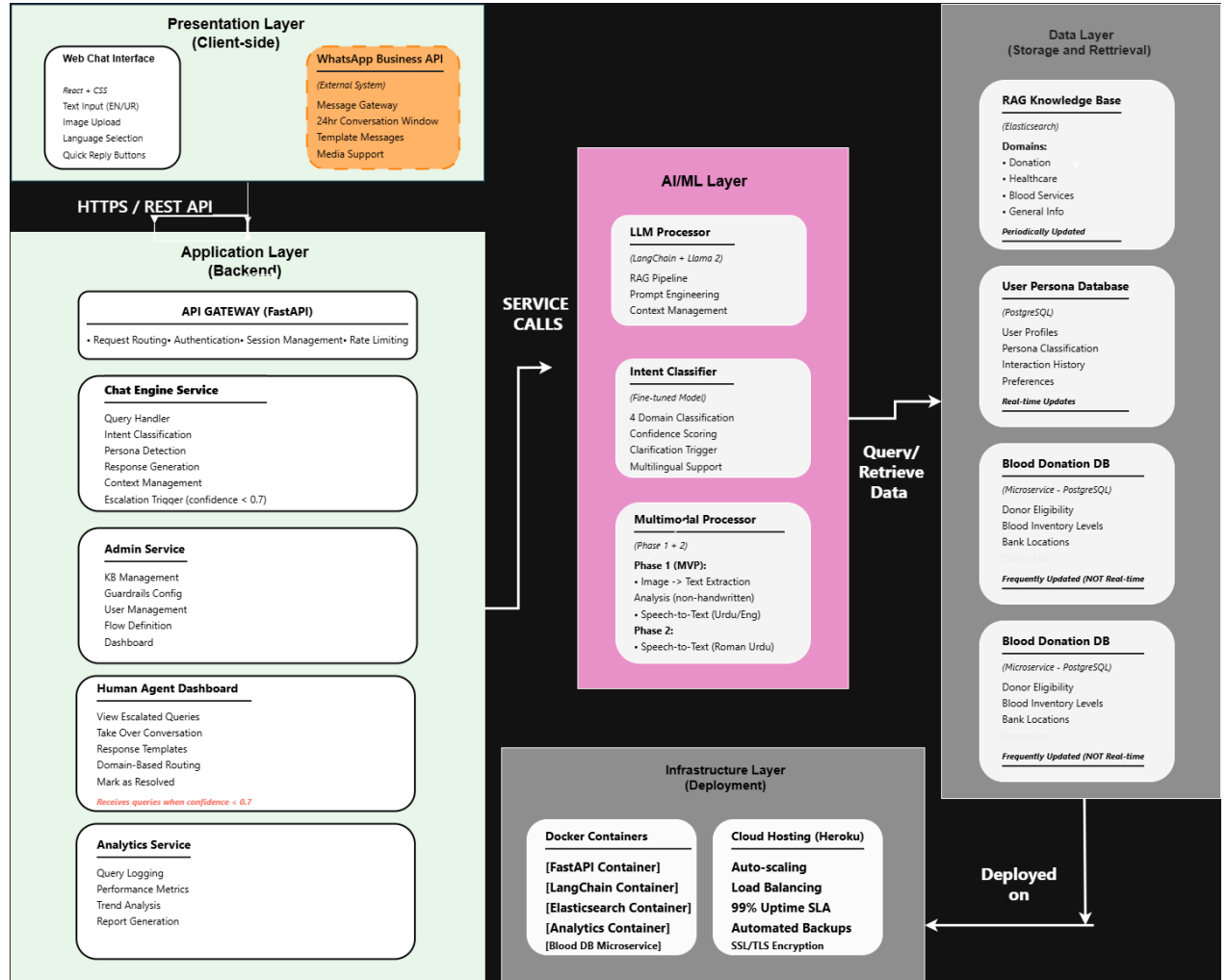


Figure 2.2: High-level System Architecture

To view the diagram clearly, kindly access the following link.

The proposed system follows a five-layer architecture: Presentation Layer, Application Layer, AI/ML Layer, Data Layer, and Infrastructure Layer. Each layer plays a distinct role in ensuring efficient communication, scalability, and performance.

2.4.1 Presentation Layer (Client-Side)

Components:

- **Web Chat Interface:** Built using React and CSS, enabling the user to interact with the system through a web interface. It supports features like text input (EN/UR), image uploads, language selection, and quick reply buttons.
- **WhatsApp Business API:** Used for external messaging functionalities such as managing conversations within a 24-hour window, sending template messages, and supporting media in chats.

Role: This layer is responsible for presenting the interface to the user, collecting user inputs, and initiating requests to the application layer via HTTP/REST APIs.

2.4.2 Application Layer (Backend)

Components:

- **API Gateway:** Manages incoming requests, handles authentication, session management, and rate limiting.
- **Chat Engine Service:** Handles the main logic for query processing, intent classification, persona detection, and response generation. It also includes context management and escalation triggers for low-confidence queries.
- **Admin Service:** Provides tools for managing the knowledge base, user roles, and defining conversation flows.
- **Human Agent Dashboard:** Assists human agents in handling escalated queries, managing conversations, and routing them based on specific domains.
- **Analytics Service:** Logs queries, monitors performance metrics, and generates reports on system usage and trends.

Role: This layer is where the core processing happens. It handles requests from the presentation layer, processes them using the chat engine, and provides feedback to the user.

2.4.3 AI/ML Layer

Components:

- **LLM Processor:** Processes the natural language input using large language models (LLMs) such as LangChain and Llama 2. It performs tasks like prompt engineering and context management.
- **Intent Classifier:** Classifies user inputs into predefined domains, scores confidence levels, and triggers escalation when confidence is low.
- **Multimodal Processor:** Handles input types like images or non-handwritten text and processes them into usable information, including support for Urdu and Roman Urdu.

Role: The AI/ML layer enhances the system's ability to process and understand user queries, making it capable of handling diverse input types and languages while maintaining context for more accurate responses.

2.4.4 Data Layer (Storage and Retrieval)

Components:

- **RAG Knowledge Base:** Stores structured data related to donation, health-care, and blood services. This database is periodically updated.
- **User Persona Database:** Tracks user profiles, including interaction history, preferences, and persona classification, to provide personalized responses.
- **Blood Donation Database:** Contains critical data on donor eligibility, blood inventory levels, and bank locations. This database is frequently updated.

Role: The data layer ensures that the necessary information is available for the application layer to process and retrieve during interactions. It stores both real-time and frequently updated data for accurate responses.

2.4.5 Infrastructure Layer (Deployment)

Components:

- **Docker Containers:** Each component, including the FastAPI service, LangChain, Elasticsearch, and analytics, is containerized to ensure consistent and scalable deployment.

- **Cloud Hosting (Heroku):** The system is hosted on Heroku with auto-scaling, load balancing, and high availability (99% uptime SLA). SSL/TLS encryption ensures secure communication.

Role: The infrastructure layer ensures that the system is deployed efficiently, scalable, and resilient. It handles the deployment of services and ensures they are readily available to process requests.

2.4.6 Relationships Between Layers

- User interaction starts at the **Presentation Layer**, where the user inputs queries.
- Requests are routed via HTTPS/REST APIs to the **Application Layer** for processing.
- The **Chat Engine Service** in the Application Layer utilizes the **AI/ML Layer** to classify intents and process natural language queries.
- **Data Retrieval** from the Data Layer occurs through queries made by application services, which fetch relevant data from the knowledge base and databases.
- The system is deployed on the **Infrastructure Layer**, ensuring scalability and security.

2.4.7 Flow of Interaction

1. **User Interaction** → Presentation Layer → API Gateway
2. **Request Routing** → Chat Engine Service → AI/ML Layer (Intent Classification & Response Generation)
3. **Data Retrieval** → Data Layer (RAG Knowledge Base, User Persona, and Blood Donation DB)
4. **Response Sent** → Presentation Layer → User (via WhatsApp or Web Chat)

This architecture ensures that each layer is responsible for a specific aspect of the system, from front-end interaction to backend processing, AI-based intent classification, and data management, all underpinned by scalable and secure infrastructure.

2.5 External Interfaces

2.5.1 User Interfaces - Wireframes

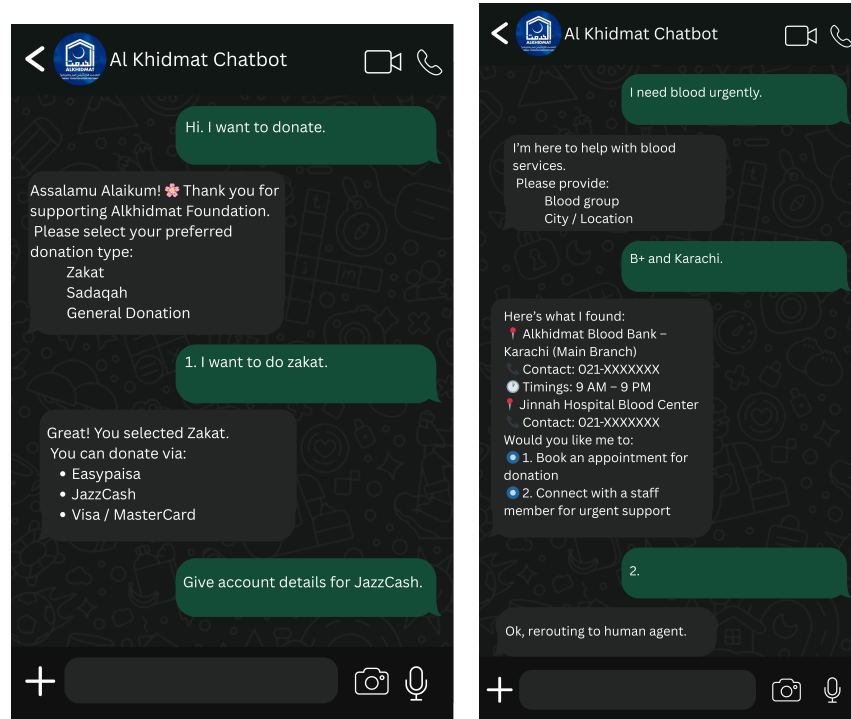


Figure 2.3: User Interface using Whatsapp

The two wireframes show how users might interact with the AI chatbot. Here, two use-cases are shown; monetary donors can select between Zakat, Sadaqah, or General Donations using quick buttons, receive verified payment options (Easypaisa, JazzCash, Visa/MasterCard), and obtain secure links with automated receipts, while complex or high-value cases are routed to human donation officers. Similarly, users seeking blood support can provide their blood group and city to instantly retrieve nearby Alkhidmat blood bank details and contact numbers; for urgent or unclear cases, the chatbot escalates the conversation to a live human agent within the same WhatsApp thread as we see in below wireframes. Both flows ensure verified information, user-friendly navigation, and human oversight for transparency and reliability.

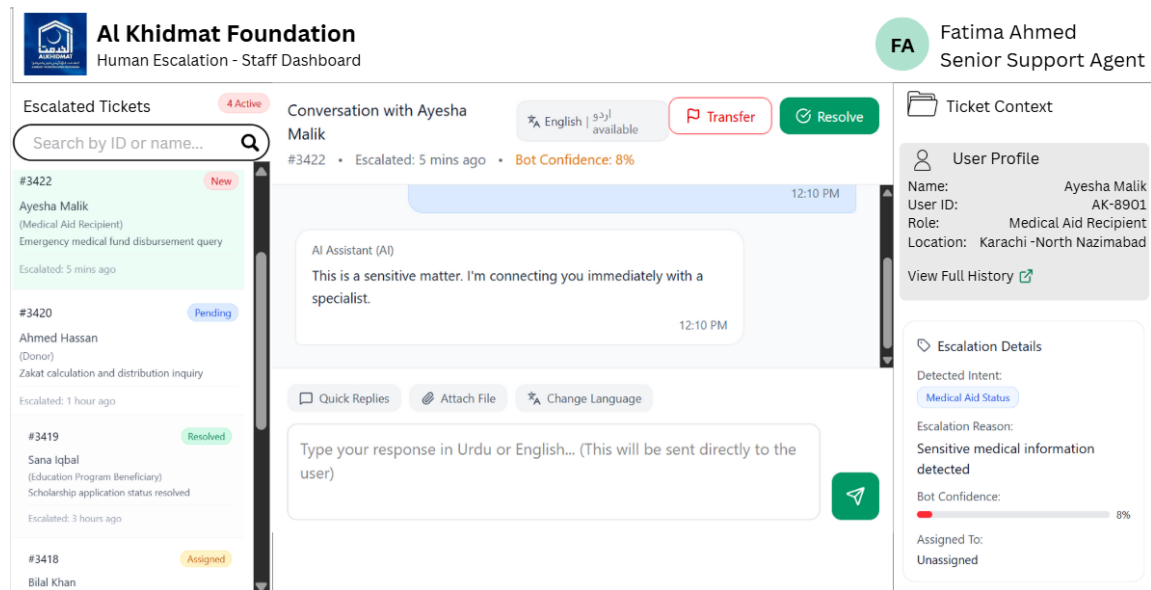


Figure 2.4: Staff Dashboard

If the AI bot is unable to confidently resolve any query, it will be escalated to Al-Khidmat’s staff. The following is the dashboard where they will receive the queries. The following example shows that the AI bit resolved a query with 8% confidence, so it has been escalated to Fatima Ahmed, a human agent. She sees that there is a new ticket. She reviews the context where the user asked sensitive information and would personally guide her - all while updating the knowledge base for future training. The dashboard includes:

- Left Panel: Live ticket queue with 4 realistic escalation scenarios (medical emergency, zakat calculation, scholarship, etc)
- Center Panel: Bilingual conversation thread showing bot-to-human handoff, where the agent has an option to transfer the query to another agent and resolve it once completed.
- Right Panel: User profile, escalation metadata, and knowledge base update form

The screenshot shows a light green rounded rectangular panel titled "Update Knowledge Base" with a book icon. It contains three sections: "Record ID:" with a text input field containing "Medical Emergency - March 2024"; "Current Status:" with a dropdown menu showing "Waiting for CNIC Verification" and a downward arrow; and "Agent Notes:" with a text area containing the placeholder text "Add notes for future reference or training data...". At the bottom is a dark green button with a floppy disk icon and the text "Save Updates".

Figure 2.5: Further Contents in Ticket Content Panel

The screenshot shows a light gray rounded rectangular panel titled "Quick Actions". It contains three items, each with an icon and text: a clipboard icon for "Copy User Details", an email icon for "Send Email Notification", and a telephone handset icon for "Schedule Follow-up Call".

Figure 2.6: Further Contents in Ticket Content Panel

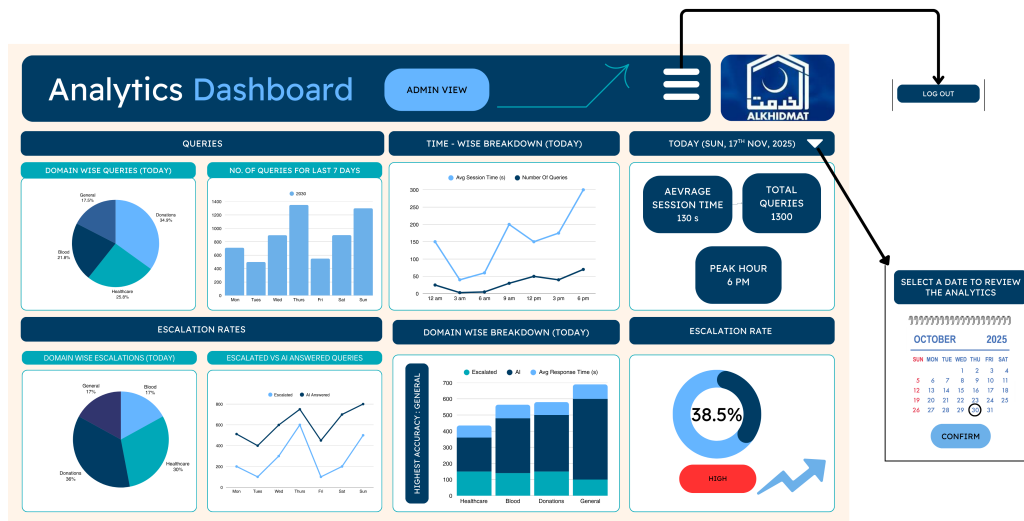


Figure 2.7: Admin's Dashboard - Analytics Monitor

The Analytics Dashboard features real-time tracking of chatbot performance and user activity, including domain-wise query distribution, daily and hourly interaction trends, and key usage metrics like total queries, average session time, and peak hours. It provides detailed insights into escalation rates, comparing AI-handled and human-assisted queries across domains to assess accuracy and efficiency. Additional features include a date selector for reviewing past analytics, visual charts for quick interpretation, and an intuitive, admin-friendly interface aligned with Alkhidmat Foundation's branding for easy monitoring and performance optimization.